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Unlocking a Secret Stash of Energy

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In this issue of *Joule*, Dahn and coworkers present the development of a hybrid lithium-ion/lithium-metal cell that allows for an extra boost of capacity when needed. This battery has the potential to help eliminate range anxiety with electric vehicles.

When I was in college, lots of my friends had VW Bugs (Beetles), one of the most popular cars for young people in those days. That's because it was cheap (especially as a used car), everyone learned how to drive a stick shift on a Bug, and it even had remarkably good (for then) fuel economy, around 20 mpg. There was a drawback for the older models, however: the fuel gauge was so inaccurate that there was always the chance you'd run out of gas. But you didn't really have to worry, because the car had a "secret" reserve gallon that would get you to the gas station or get you home. I hadn't thought of that in many years, until I read this paper from Jeff Dahn's lab and started to wonder, what if electric cars had a secret stash of energy that provided an extra 50 miles of range?

The lithium battery world has long been focused on increasing energy density, but for some applications (cars, cell phones), gravimetric energy density—the focus of most academic work—is of secondary importance in comparison with volumetric energy density.

Consider a Tesla Model 3 with a 74-kWh battery pack made from Tesla/Panasonic 2170 cells with 246 Wh/kg and 711 Wh/l.¹ A 10% reduction in battery weight would save 30 kg, considerably less than 1% of the weight of the car. As such, the effect on range might well be too small to affect the value of the car. On the other hand, a 10% reduction in battery volume would save 10.5 L. This extra space could be used to enhance passenger comfort by providing additional knee and head room by enough to be noticed and appreciated. Furthermore, the volume reduction could make packaging the car mechanicals easier for the manufacturer, which, in turn, could save cost and weight.

The present work, which can be read in this issue of *Joule*, focuses on the volumetric energy density improvement possible from creating a hybrid Li-ion/Li-metal battery.² The improvement comes from borrowing an idea already in use but applied here in an interesting and novel way. In some current batteries, small amounts of Si are added to

graphite anodes to improve energy density. Although Si typically reduces the life of batteries, this effect can be mitigated when combined with graphite, because the delithiation potential of Si is higher than that of graphite. As a result, typical discharge cycles might not access the lithiated Si, reducing its effect on life while still providing extra energy when needed. The authors ask, can an analogous benefit be achieved by plating Li metal on graphite, providing extra energy—but only when needed? What impresses me about this work is that we are all taught from diapers that plating Li metal on graphite is a terrible thing. It is never easy to think outside the box and go against what we all "know," but these authors did.

The authors built cells with a substantial excess cathode capacity, reversing the usual situation, in which anode capacity is larger in order to reduce the possibility of plating. For this "hybrid cell," no plating occurred up to a charging voltage of 4.0 V, at which point the graphite was fully lithiated. Charging between 4.0 and 4.4 V accessed the remaining cathode Li, depositing it on the graphite anode as metallic Li. The result was a 20% to 25% improvement in volumetric energy density when charged to

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4.4 V, in comparison with that in a conventional cell with the same cathode capacity. This improvement came partly because the volume of graphite was reduced and partly because higher voltages were accessed when using plated Li. Of course, there was a price to pay, a drastic reduction in life, to below 100 cycles. So, the authors asked, can we salvage the benefit of higher volumetric energy density if we only plate Li on 1 cycle out of 10?

The authors offer the following scenario. Consider a vehicle that has a range of 400 km with a conventional cell and compare that to a vehicle with a hybrid cell with the same amount of cathode. When charged to 4.0 V ("lithium ion mode"), the hybrid cell delivers only 300 km, although, because there is no plating, the cell life is not affected. But this vehicle would have a range of 480 km when operated in hybrid mode ("boosted cycles"), when charged to 4.4 V. Remarkably, if there is only 1 boosted charge out of 10, the authors obtain fairly long lifetimes for their hybrid cells, with a capacity drop of only about 10% after 400 cycles. This is far better than the lifetimes observed with so-called "anode-free" cells. As long as the vehicle owner is usually satisfied with a 300 km range, the owner can get up to 480 km range, occasionally. The authors suggest—but do not demonstrate—that the limiting factor is the total number of boosted cycles, not the fact that each boosted cycle was followed by 9 cycles charged to 4.0 V. If the authors are correct here, then several boosted cycles in a row could allow for taking long trips using only boosted cycles, with a 480 km range, without paying too high a durability penalty.

There is next the question of whether this hybrid operation increases the chances for a short circuit and fire because of dendrite growth on the plated Li. The authors show that this possibility can be reduced by applying

a stack pressure of 1.2 MPa on the cells. Such a pressure could be developed and maintained by, for example, a screw and a lock washer. As was shown previously in their lab,^{2,3} the result is a Li-metal morphology with smoother, flatter, and larger grains and with dendrite penetration substantially inhibited.

The pressure results lead to an interesting question: how and why does application of this relatively modest pressure inhibit dendrite growth and penetration? Conventionally, the mechanics of Li dendrites are understood through the model of Monroe and Newman^{4–6} and followers,⁷ in which a Li-metal protrusion presses conformally against a polymer separator over a spatial region containing just that single protrusion (proto-dendrite), say several square microns. The model predicts that a separator shear strength about double that of Li metal, greater than 4 GPa, is necessary to force Li to grow flat and prevent dendrite penetration. Yet, in the present work, a soft polymer separator was able to accomplish the same thing at a stack pressure of only 1.2 MPa.

The fact that these results are not predicted by the Monroe-Newman model led us to suggest that an important assumption of the model does not apply in Li-metal/liquid-electrolyte systems, such as that studied in the present work: namely, the assumption that contact between the Li metal and the separator is (nearly) conformal. To support that suggestion, we developed a 3D electrochemical/contact mechanics model that considered Li metal as a rough, elastic-perfectly-plastic material with creep.⁸ The model analyzed the effects of local contact interactions on Li plating/stripping over a 400-micron $\times$ 400-micron area. Our results showed that, far from conformal contact between Li metal and the separator, all of the load is carried at just the peaks of the tallest Li asperities (proto-dendrites), where stresses

cannot exceed three times the yield strength (the hardness) of Li metal, whereas most of the Li surface is not in contact with the separator at all. Under these conditions, the primary effect of an external load is creep:⁹ Li protrusions "melt away," under local stresses as small as a few MPa, leading to the flat morphology seen in the present paper. The effects of creep, which extend over hundreds of square microns, cannot be captured by a single-dendrite-scale analysis.

Finally, one can ask whether hybrid cells as described in this paper can actually be practical, assuming that high pressure solves the dendrite penetration problem. First, we note that there is a difference between the boost provided by hybrid cells and the "secret stash" in the old VW Bug. In the latter case, the stash was an emergency backup system, to be used when you would otherwise be stranded for lack of gasoline. In contrast, in order to get a boost from the hybrid cell, you need to charge to 4.4 V ahead of time.

So, how do people actually use their cars? The paper points out that only 1% of automotive trips are greater than 300 km.¹⁰ If we can live with 300 km almost all of the time, how valuable is a boost to 480 km for 10% of the charges? Perhaps not very valuable by itself: compared to a vehicle with a range of 400 km, it would be rare to need the extra 80 km, especially because the boost can only be used in 1 charge out of 10. Furthermore, the market (at least in the US) has shown that people tend to demand a greater range, whether or not they actually use it with any regularity. However, the true value of the hybrid cell is its improved volumetric efficiency. How many people would take a 300 km range instead of a 400 km range if in return they got not only more interior room/comfort but also an ability to sometimes get a 480 km range? That's a question for marketing people, not

necessarily battery scientists, but I would guess that the answer is at least “some,” if it turns out that it is possible to boost several consecutive cycles without compromising life.

In the end, I think that this paper can be viewed as an excellent example of how a simple idea, requiring no new technology, can break conventional paradigms to produce a cell that, at least plausibly, can be commercialized.

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Toward Sustainable H₂ Production: Linking Hydrogenase with Photosynthesis

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Molecular hydrogen, H₂, is an energy carrier that is increasing in popularity as an alternative fuel. Presently, the major part of commercially available H₂ is produced from fossil resources. To make H₂ a true contender to fossil fuels, a sustainable production via water splitting by renewable electricity in combination with earth-abundant catalysts or by photosynthetic microorganisms is required.

Biological H₂ production usually involves enzymes known as hydrogenases that are found in microorganisms. The majority of hydrogenases can be classified into two groups based on their active site: the NiFe hydrogenases that have a heterometallic active site with one Fe and one Ni ion and the FeFe hydrogenases, carrying an inorganic Fe₂ cofactor in the active site. Interestingly, each species of microor-

ganism has either one or the other type, but never both.

In existing biogas plants, biomass—usually waste—is fermented by microorganisms to methane, CO₂ and H₂. Since biomass is produced by photosynthesis, this process indirectly utilizes solar energy. Many photosynthetic microorganisms carry hydrogenases and thus have the potential to produce H₂

directly from solar energy in a cleaner, more efficient way than by fermentation. This was described first by Gaffron, who discovered H₂ production by green algae in 1937.¹ Ever since, it has been a dream to create a direct link between photosynthesis and hydrogenases to drive solar H₂ production from water (reaction 1 in [Scheme 1](#) and [Figure 1](#)).

Oxygenic photosynthesis is powered by light-driven charge separations in two reaction centers ([Figure 1](#)). Photosystem II (PSII) oxidizes water and reduces the plastoquinone (PQ) pool while releasing molecular oxygen and protons (reaction 2). The electrons are delivered to photosystem I (PSI) via

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